交通大数据

对数几率回归

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❑ 学习目标

- 二项 Logistic 回归
- 多项 Logistic 回归

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二项 Logistic 回归

广义线性模型

- 概率密度或质量函数如下

$$f(y;\theta,\phi)=\exp\left(\frac{y\theta-b(\theta)}{a(\phi)}+c(y,\phi)\right)$$

- 二项 Logistic 回归实际是0-1分类问题，即 $y_i \in [0,1]$
- 针对0-1分类问题，广义线性模型的随机部分该如何选取？
- 伯努利分布！

$$f(y)=p^y(1-p)^{1-y}$$

二项 Logistic 回归

❑ 广义线性模型

- 概率质量函数变形如下

$$f(y;\theta,\phi)=\exp\left(\frac{y\theta-b(\theta)}{a(\phi)}+c(y,\phi)\right)$$

$$f(y)=p^y(1-p)^{1-y}$$

$$\Downarrow$$

$$\begin{aligned}f(y)&=\exp\Big(y\log p+(1-y)\log(1-p)\Big)\\&=\exp\Big(y\cdot\log\big(p/(1-p)\big)+\log(1-p)\Big)\end{aligned}$$

二项 Logistic 回归

广义线性模型

$$
\begin{aligned}
f(y) &= \exp\Big(y\log p + (1-y)\log(1-p)\Big) \\
&= \exp\Big(y\cdot\log\big(p/(1-p)\big) + \log(1-p)\Big)
\end{aligned}
$$

- 对应的广义线性模型的参数如下
  - $\boxed{\theta = \log(p/(1-p))}$，或 $p = e^{\theta}/(1+e^{\theta})$
  - $b(\theta) = -\log(1-p) = \log\left(1+e^{\theta}\right)$
  - $a(\phi) = 1$，$c(y,\phi) = 0$

二项 Logistic 回归

广义线性模型

- 二项分布的期望 = 二项 Logistic 回归的估计对象 = p
- 我们有
  - $\theta = \log(p/(1-p))$
  - 标准连接函数 $\theta = \eta = \mathbf{X}^{\mathrm{T}}\boldsymbol{\beta}$
- 因此我们有

$$\log\frac{p}{1-p} = \boxed{\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta} = \log\frac{y}{1-y}}$$

二项 Logistic 回归

广义线性模型

$$\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta} = \log\frac{y}{1-y}$$

- y 是取值为1的概率，$1-y$ 是取值为0的概率
- 两者的比值 $\frac{y}{1-y}$ 表示取值为1的**相对可能性**，即“几率”（odds）
- 对“几率”取对数，则是“对数几率”（log odds）

二项 Logistic 回归

广义线性模型

$$\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta} = \log\frac{y}{1-y}$$

- y 到底是什么形式?

$$\frac{y}{1-y} = e^{\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta}}$$

$$y = \frac{1}{1+e^{-\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta}}}$$

二项 Logistic 回归

❑ 广义线性模型

- 考虑定义 $\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta} = \eta$
- 则 $y = \frac{1}{1+e^{-\mathbf{X}^{\mathrm{T}}\boldsymbol{\beta}}} = \frac{1}{1+e^{-\eta}}$
  - 对数几率函数（logistic function）或 Sigmoid 函数
  - 平滑S型曲线（连续可微）
  - 输入$[-\infty,+\infty]$ → 输出$[0,1]$
  - $\eta = 0$ 附近变化陡峭，保证了分类在边界处的敏感性

$$y=\begin{cases}0, & z<0\\ 0.5, & z=0\\ 1, & z>0\end{cases}$$

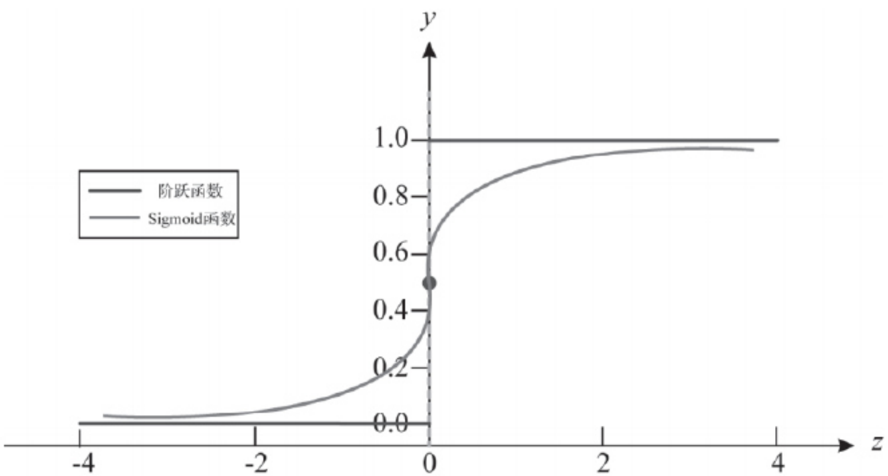


图 6-8　单位阶跃函数与 Sigmoid 函数

二项 Logistic 回归

❑ 广义线性模型

- 对比单位阶跃函数 （unit step function）
  - 不连续可微
  - 未能给分类结果一个“置信程度”
    - $\eta = 2$ 和 $\eta = 4$ 取值为1的概率均为 1，但是 $\eta = 4$ 取值为1的概率理应更大

$$y = \begin{cases} 0, & z < 0 \\ 0.5, & z = 0 \\ 1, & z > 0 \end{cases}$$

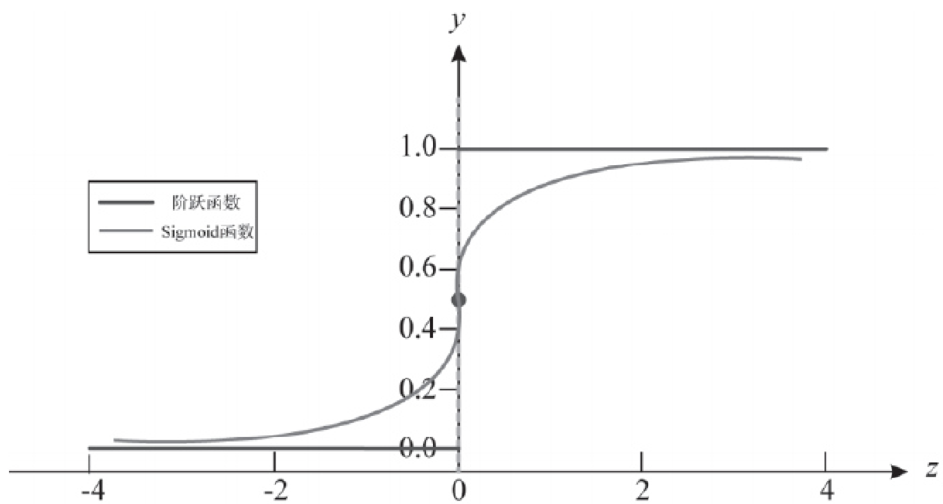


图 6-8　单位阶跃函数与 Sigmoid 函数

Logistic Regression

- Using a logistic regression model, the probability of exceeding the speed limit under certain traffic conditions can be estimated using the following equation

$$P(y_i = 1) = \frac{1}{1 + e^{-x'_i\beta}} \quad (i = 1, 2, \ldots, n)$$

- where P(yi = 1) denotes the probability of exceeding the speed limit, and $-x'_i\boldsymbol{\beta}$ is the multiple linear combination of explanatory variables:

$$x'_i\beta = \beta_0 + \beta_1 x_{1i} + \ldots + \beta_k x_{ki}$$

- This model is estimable by standard maximum likelihood methods, with the likelihood function given as:

$$L(\beta) = \prod_{i=1}^{n}\left[\left(\frac{1}{1 + e^{-x'_i\beta}}\right)^{y_i}\left(1 - \frac{1}{1 + e^{-x'_i\beta}}\right)^{(1-y_i)}\right]$$

- The interpretation of variable parameter in logistic regression:

$$\left(\frac{P_i}{1-P_i}\right) = EXP^{B_0+B_iX_i} = EXP^{B_0}EXP^{B_iX_i}$$

- The fundamental equation for the logistic regression shows that when the value of an independent variable increases by one unit, and all other variables are held constant, the new probability ratio $[P_i/(1-P_i)]$ is given as

$$\left(\frac{P_i}{1-P_i}\right)^* = EXP^{B_0}EXP^{B_i(X_i+1)} = EXP^{B_0}EXP^{B_iX_i}EXP^{B_i} = \left(\frac{P_i}{1-P_i}\right)EXP^{B_i}$$

Logistic Regression

$$\left(\frac{P_i}{1-P_i}\right) = EXP^{B_0+B_iX_i} = EXP^{B_0}EXP^{B_iX_i}$$

$$\left(\frac{P_i}{1-P_i}\right)^* = EXP^{B_0}EXP^{B_i(X_i+1)} = EXP^{B_0}EXP^{B_iX_i}EXP^{B_i} = \left(\frac{P_i}{1-P_i}\right)EXP^{B_i}$$

Thus when independent variable X_i increases by one unit, with all other factors remaining constant, the odds $[P_i/(1-P)]$ increases by a factor EXP^{B_i}. The factor EXP^{B_i} is called the odds ratio (OR) and ranges from zero to positive infinity. It indicates the relative amount by which the odds of the outcome increases (OR > 1) or decreases (OR < 1) when the value of the corresponding independent variable increases by 1 unit.

二项 Logistic 回归

- 如何定量分析自变量的变化对因变量的影响？
  - 弹性（连续变量）

$$\log\frac{p_i}{1-p_i}=\mathbf{X}_i^{\mathrm{T}}\boldsymbol{\beta} \rightarrow p_i=\frac{1}{1+e^{-\mathbf{X}_i^{\mathrm{T}}\boldsymbol{\beta}}}$$

$$E_{x_{ik}}^{p_i}=\frac{\Delta p_i}{p_i}\Big/\frac{\Delta x_{ik}}{x_{ik}}=\frac{\Delta p_i}{\Delta x_{ik}}\frac{x_{ik}}{p_i}\approx\frac{\partial p_i}{\partial x_{ik}}\frac{x_{ik}}{p_i}$$

$$\Downarrow$$

$$E_{x_{ik}}^{p_i}=p_i(1-p_i)\beta_{ik}\frac{x_{ik}}{p_i}=(1-p_i)\beta_{ik}x_{ik}$$

二项 Logistic 回归

- 如何定量分析自变量的变化对因变量的影响？
  - 伪弹性（名义尺度变量）

$$E_{x_{ik}}^{p_i} = \frac{\Delta p_i}{p_i} \Big/ \frac{\Delta x_{ik}}{x_{ik}} = -\frac{\Delta p_i}{p_i}$$

$$= -\frac{\dfrac{1}{1+e^{-(x_1\beta_1+\cdots+x_{k-1}\beta_{k-1}+x_{k+1}\beta_{k+1}+\cdots)}} - \dfrac{1}{1+e^{-(x_1\beta_1+\cdots+\beta_k+\cdots)}}}{\dfrac{1}{1+e^{-(x_1\beta_1+\cdots+\beta_k+\cdots)}}}$$

$$= \frac{e^{\beta_k}-1}{e^{\beta_k}\left(1+e^{(x_1\beta_1+\cdots+x_{k-1}\beta_{k-1}+x_{k+1}\beta_{k+1}+\cdots)}\right)}$$

- The likelihood ratio test statistic is

$$X^2 = -2[LL(\beta_R) - LL(\beta_u)]$$

where $LL(\beta_R)$ is the log likelihood at convergence of the "restricted" model (sometimes considered to have all parameters in β equal to 0, or just to include the constant term, to test overall fit of the model), and $LL(\beta_U)$ is the log likelihood at convergence of the unrestricted model.

- The χ^2 statistic is χ^2 distributed with the degrees of freedom equal to the difference in the numbers of parameters in the restricted and unrestricted model (the difference in the number of parameters in the β_R and the β_U parameter vectors.

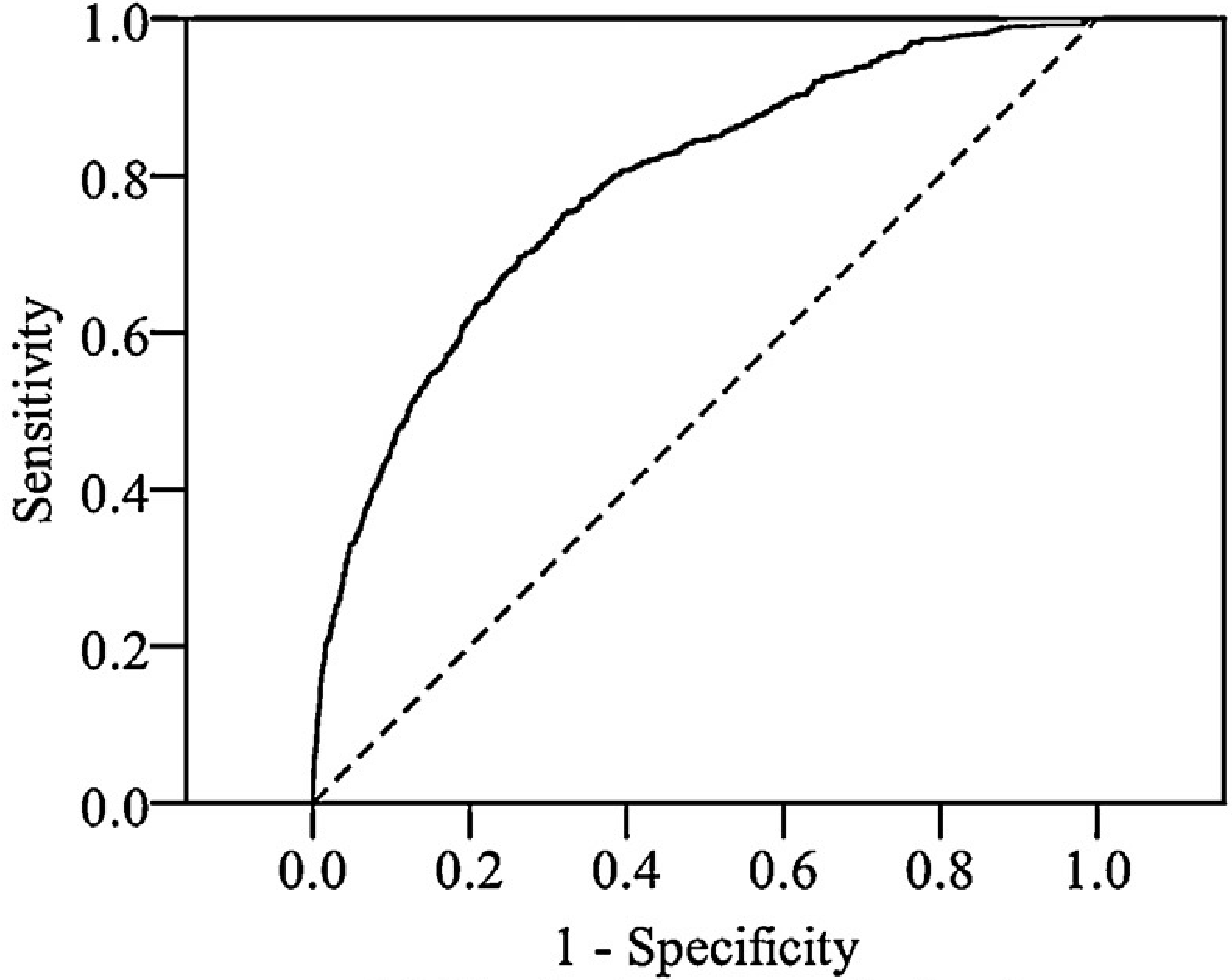
- Another measure of overall model fit is the ρ^2 statistic. The ρ^2 statistic is

$$\rho^2 = 1 - \frac{LL(\beta)}{LL(0)}$$

where $LL(\beta)$ is the log likelihood at convergence with parameter vector β and $LL(0)$ is the initial log likelihood (with all parameters set to zero).

- The perfect model would have a likelihood function equal to one (all selected alternative outcomes would be predicted by the model with probability one, and the product of these across the observations would also be one) and the log likelihood would be zero, giving a ρ^2 of one.
- Thus the ρ^2 statistic is between zero and one and the closer it is to one, the more variance the estimated model is explaining.

ROC Curve



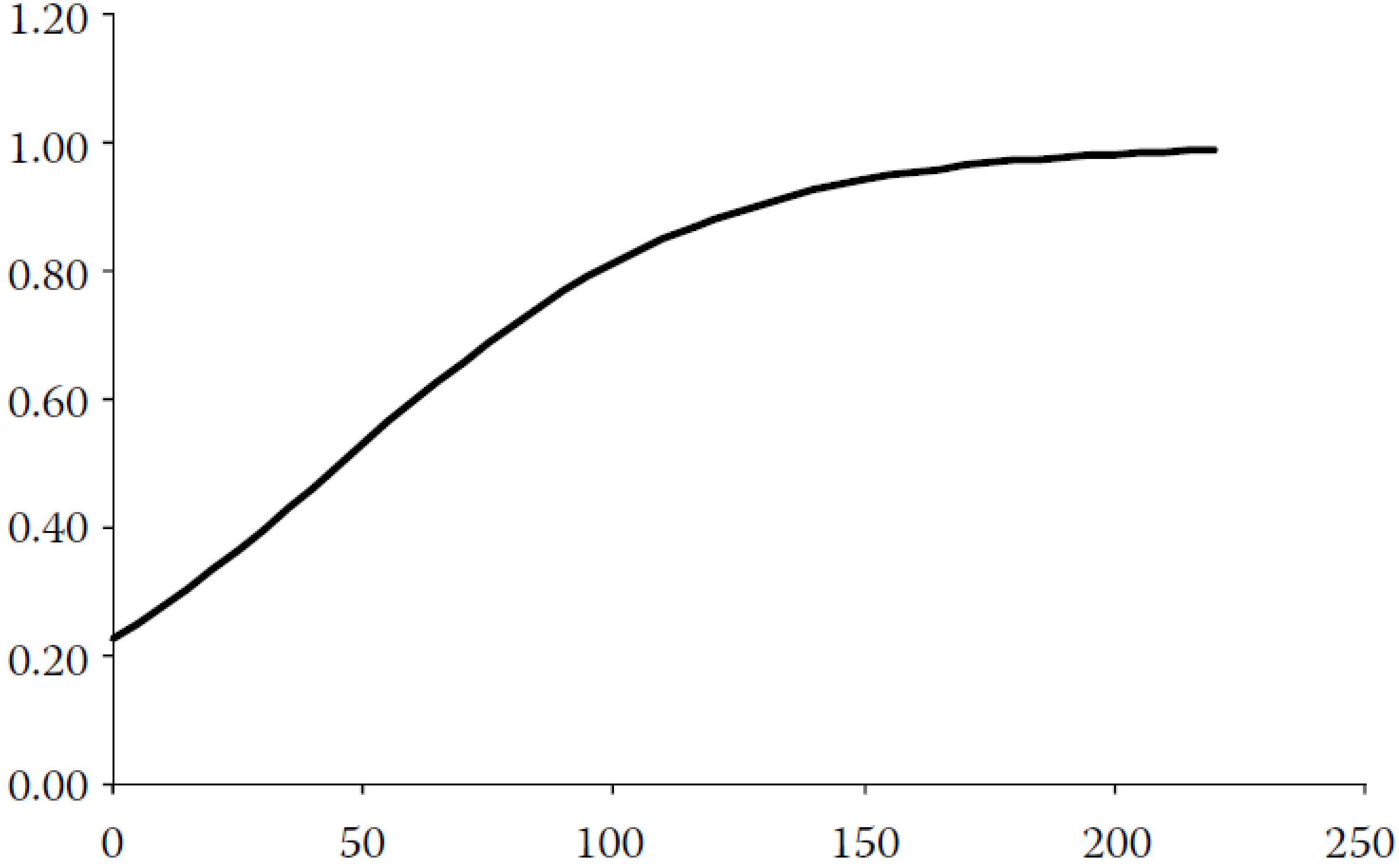
Logistic Regression

Variable No.	Variable Description
1	Indicator variable for primary incident (0 = not linked to secondary incident; 1 = linked to secondary incident)
2	Clearance time of incident in minutes
3	Indicator variables for season of year (fall, winter, spring, summer)
4	Indicator variables for vehicle type (car, truck, semi, van, bus)
5	Road position at time of incident (left lane, right lane, right shoulder, and ramp)
6	Indicator variable for weekend travel
7	Indicator variable for AM or PM peak period travel

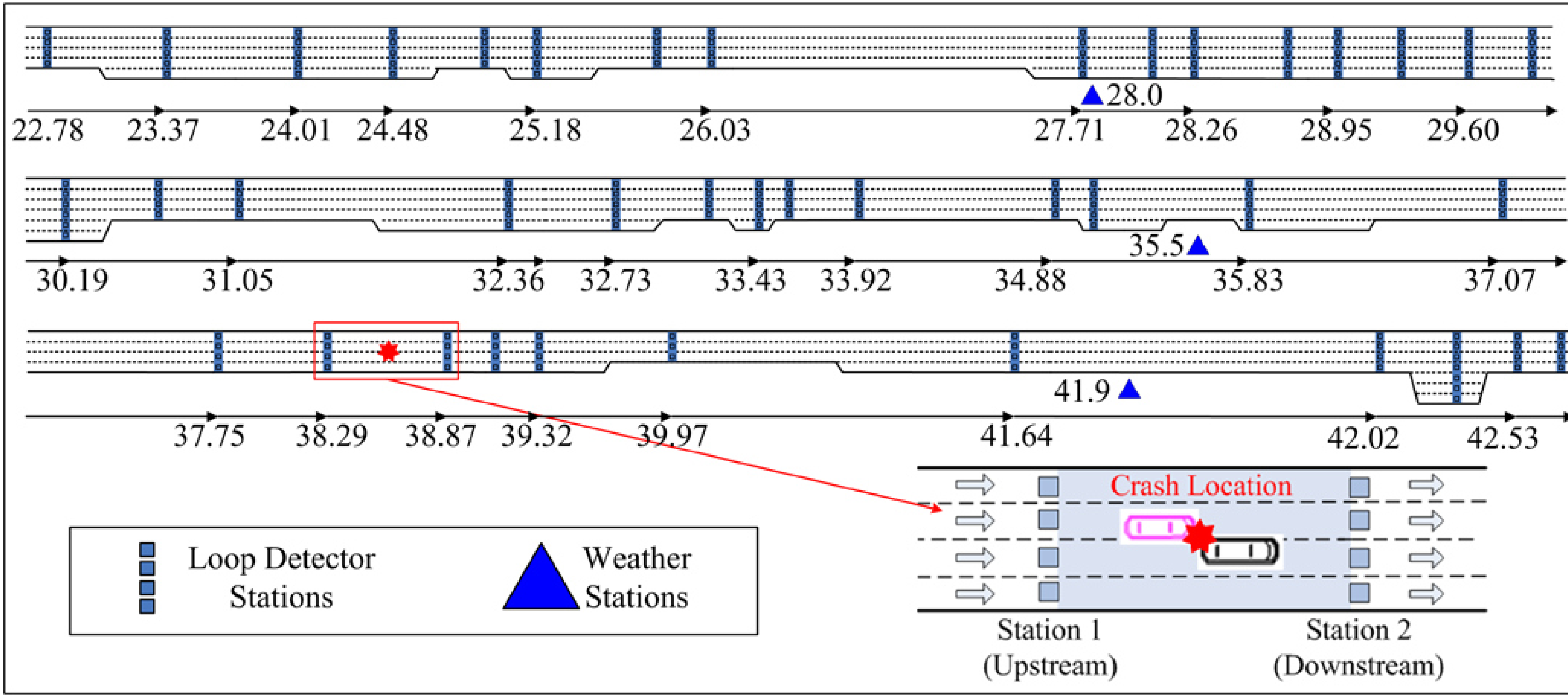
Logistic Regression

Variable	Parameter Estimate	t-statistic	$P > \|z\|$	95% Coefficient Interval
Constant	–1.22	–8.23	.000	–1.516, –0.933
Clearance time of incident in minutes	0.027	6.35	.000	0.0187, 0.0354
Winter indicator variable (1 if incident occurred in the winter, 0 otherwise)	–0.417	–2.23	.026	–0.783, –0.051
Van indicator (1 if vehicle is a van, 0 otherwise)	–1.848	–2.48	.013	–3.309, –0.386
Number of observations	741			
Log likelihood	–446.64			
Likelihood ratio chi-square (3 d.f.)	63.30			
Probability > chi-square	<.0001			
Pseudo R-squared	0.0662			

Logistic Regression



Logistic Regression



Logistic Regression

Symbol	Variables
VehCnt_u	Average 30-second vehicle count at the upstream station (veh/30 s)
DetOcc_u	Average 30-second detector occupancy at the upstream station (%)
AvgSpd_u	Average 30-second speed at the upstream station (mile/h)
OccDev_u	Std. dev. of 30-second detector occupancies at the upstream station (%)
SpdDev_u	Std. dev. of 30-second mean speeds at the upstream station (mile/h)
CvSpd_u	Coefficient of variation of 30-second mean speeds at the upstream station (mile/h)
OccDif_u	Average absolute difference in 30-second detector occupancies between adjacent lanes at the upstream station (%)
SpdDif_u	Average absolute difference in 30-second mean speeds between adjacent lanes at the upstream station (mile/h)
VehCnt_d	Average 30-second vehicle counts at the downstream station (veh/30s)
DetOcc_d	Average 30-second detector occupancy at the downstream station (%)
AvgSpd_d	Average 30-second speed at the downstream station (mile/h)
OccDev_d	Std. dev. of 30-second detector occupancies at the downstream station (%)
SpdDev_d	Std. dev. of 30-second mean speeds at the downstream station (mile/h)
CvSpd_d	Coefficient of variation of 30-second mean speeds at the downstream station (mile/h)
OccDif_d	Average absolute difference in 30-second detector occupancies between adjacent lanes at the downstream station (%)
SpdDif_d	Average absolute difference in 30-second mean speeds between adjacent lanes at the downstream station (mile/h)
$\text{AvgCnt}_{u\text{-}d}$	Average absolute difference in vehicle counts between upstream and downstream stations (veh/30s)

Logistic Regression

$AvgCnt_{u\text{-}d}$	Average absolute difference in vehicle counts between upstream and downstream stations (veh/30s)
$AvgOcc_{u\text{-}d}$	Average absolute difference in detector occupancies between upstream and downstream stations (%)
$AvgSpd_{u\text{-}d}$	Average absolute difference in speeds between upstream and downstream stations (mile/h)
$DevCnt_{u\text{-}d}$	Std. dev. of absolute difference in vehicle counts between upstream and downstream stations (veh/30s)
$DevOcc_{u\text{-}d}$	Std. dev. of absolute difference in detector occupancies between upstream and downstream stations (%)
$DevSpd_{u\text{-}d}$	Std. dev. of absolute difference in speeds between upstream and downstream stations (mile/h)
$DetDist_{u\text{-}d}$	Distance between upstream and downstream stations (mile)
$Width_s$	Road surface width (ft)
$Width_o$	1 = if outer Shoulder width > 10ft; 0 = otherwise
$Width_i$	1 = if inner Shoulder width > 10ft; 0 = otherwise
$Width_m$	Inner median width (ft)
Lanes	Number of lanes
On-ramp	1 = if there is an on-ramp between upstream and downstream stations; 0 = otherwise
Off-ramp	1 = if there is an off-ramp between upstream and downstream stations; 0 = otherwise
Curve	1 = Curve section; 0 = otherwise
Peak	1= Peak period; 0 = otherwise
Weather	1 = adverse weather conditions(rain or fog); 0 = otherwise

Logistic Regression

Crash (KA, BC, and PDO) vs. Non-crash					
Parameter	Estimate	Std. Error	Wald χ^2	Pr>Chisq	Elasticity
$DetOcc_u$	0.069	0.007	103.180	<.0001	0.484 (0.353[b])
$SpdDev_u$	0.046	0.016	8.510	0.004	0.175 (0.11)
$SpdDev_d$	0.047	0.017	7.615	0.006	0.177 (0.106)
$OccDif_d$	0.093	0.011	71.219	<.0001	0.271 (0.282)
$DetDist_{u-d}$	0.770	0.090	73.463	<.0001	0.403 (0.296)
$Width_s$	-0.040	0.008	25.216	<.0001	-2.042 (0.223)
$Width_o$	-0.683	0.150	20.718	<.0001	-0.336 (0.021)
Curve	0.352	0.121	8.521	0.004	0.171 (0.01)
Intercept	-1.979 (-5.397[a])	0.427	21.512	<.0001	—

Summary statistics:

$-2LL(c) = 5292.567; -2LL(\beta) = 4760.989$

$-2[LL(c) - LL(\beta)] = 531.578$ (8 df); $P < 0.0001$

对数几率回归

❑ 学习目标

- 二项 Logistic 回归
- 多项 Logistic 回归

多项 Logistic 回归

- 如何从二分类转化为多分类？思路一
  - 对多分类问题进行拆分，变成多个二分类问题
  - 即构建多个二项 Logistic 回归
    - 假设我们的数据集有 k 类
    - 挑选出类别为 a 的样本，$a = 1,2, \dots, k$，该样本记为1
    - 剩余 $k - 1$ 类的样本记为0
    - 运用二项 Logistic 回归 k 次，构建 k 个二分类器
    - 对某个测试样本，其类别对应 k 个二分类器中输出值最大的

多项 Logistic 回归

- 如何从二分类转化为多分类？思路二
  - 用 Softmax 函数替换对数几率函数

$$h_\theta(X_i) := \frac{1}{1 + e^{-\theta^\top X_i}}$$

$$h_\theta(\mathrm{X}) = \begin{pmatrix} p(y=1 \mid \mathrm{X};\theta) \\ p(y=2 \mid \mathrm{X};\theta) \\ \vdots \\ p(y=k \mid \mathrm{X};\theta) \end{pmatrix} = \frac{1}{\sum_{i=1}^{k} e^{\theta_i^\top \mathrm{X}}} \begin{pmatrix} e^{\theta_1^\top \mathrm{X}} \\ e^{\theta_2^\top \mathrm{X}} \\ \vdots \\ e^{\theta_k^\top \mathrm{X}} \end{pmatrix}$$

  - Softmax 函数推广了对数几率函数

多项 Logistic 回归

❑ 多个二分类器 vs. Softmax 分类器

- 取决于样本数据中各类别之间是否互斥
  - 若各类别之间明显互斥 → Softmax 分类器
  - 若各类别之间有交叉 → 多个二分类器